# Self Driving Car

A Minor Project

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# Why Self Driving Car

- Human drivers make errors
- Self Driving Technology is a promising direction
- Home-grown technology is crucial
- Inspired by our seniors' work (SDC Major Project)
- Grow our skills in robotics

# Scope

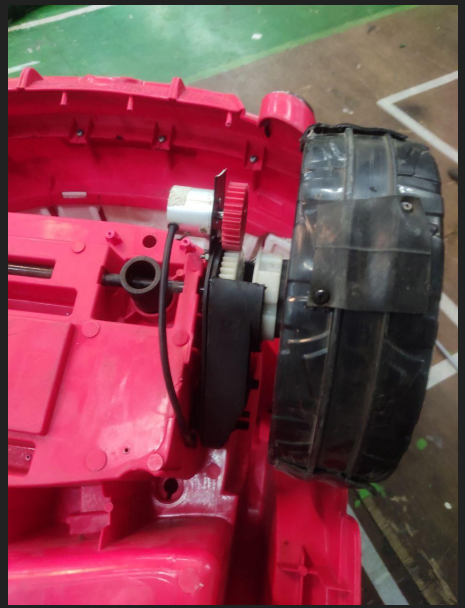
- Autonomous operation in a small scale environment
- Use ROS2 for middleware
- Reuse components from Nav2 and ros2_control
- Target slow speed operation in good lighting conditions



# Our Project

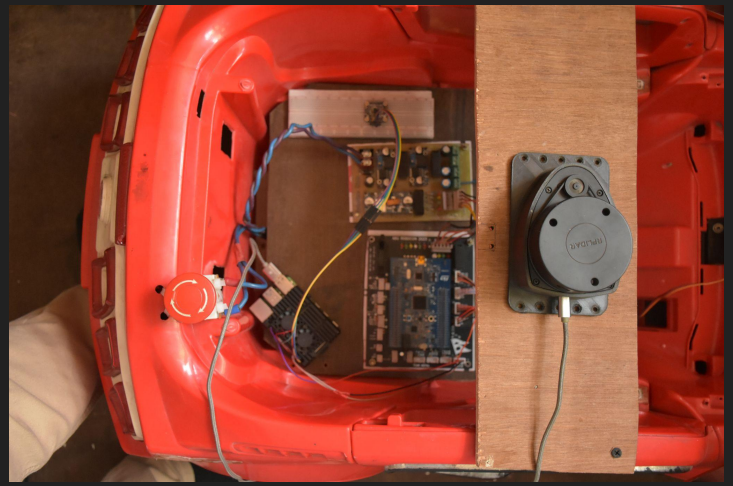
# Drivetrain Repair

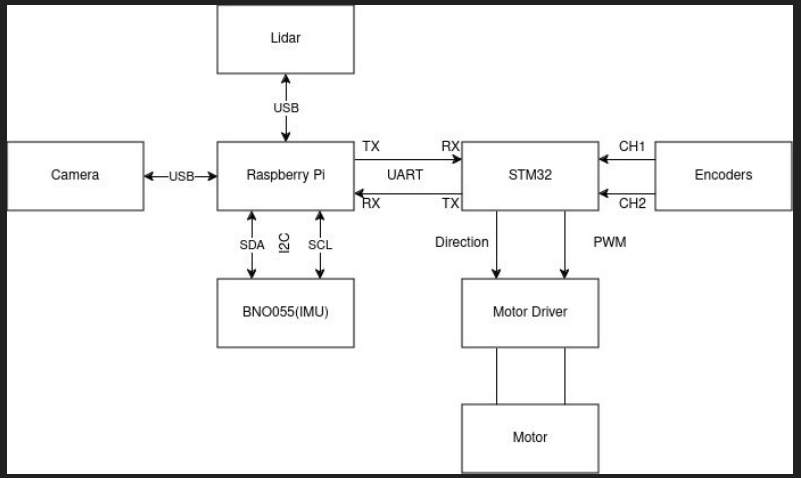




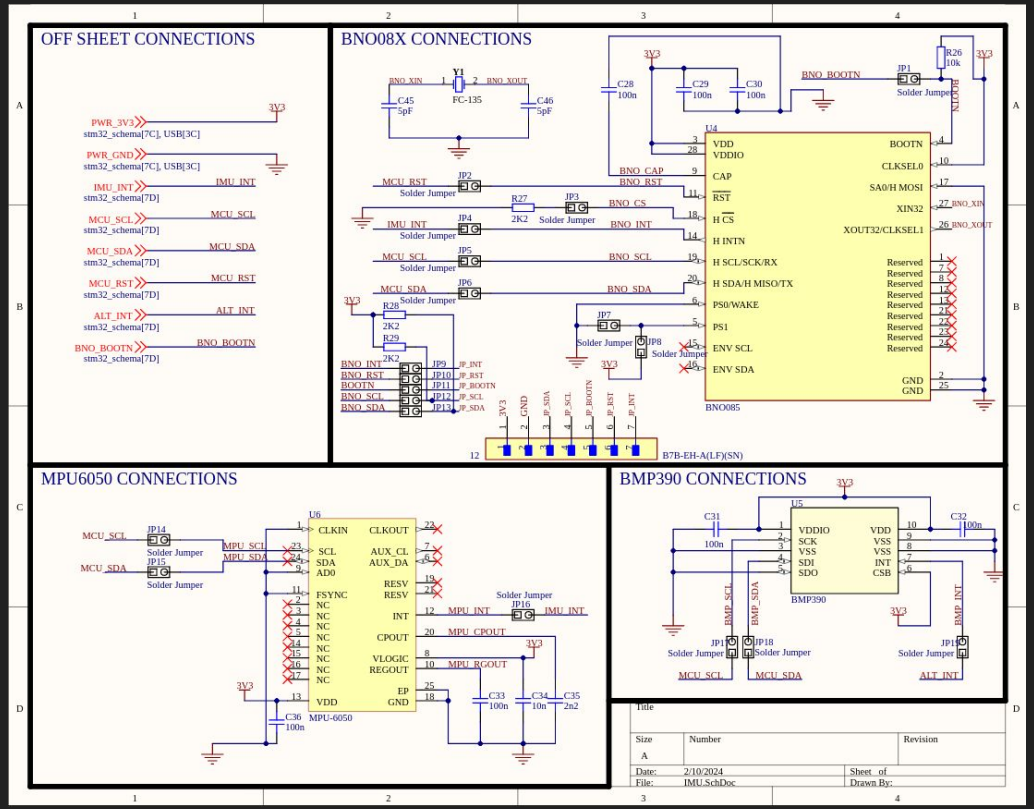


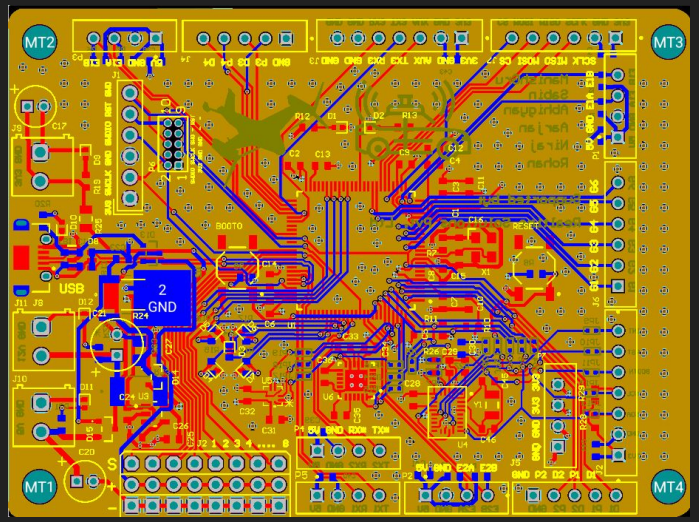
# Electronics Setup



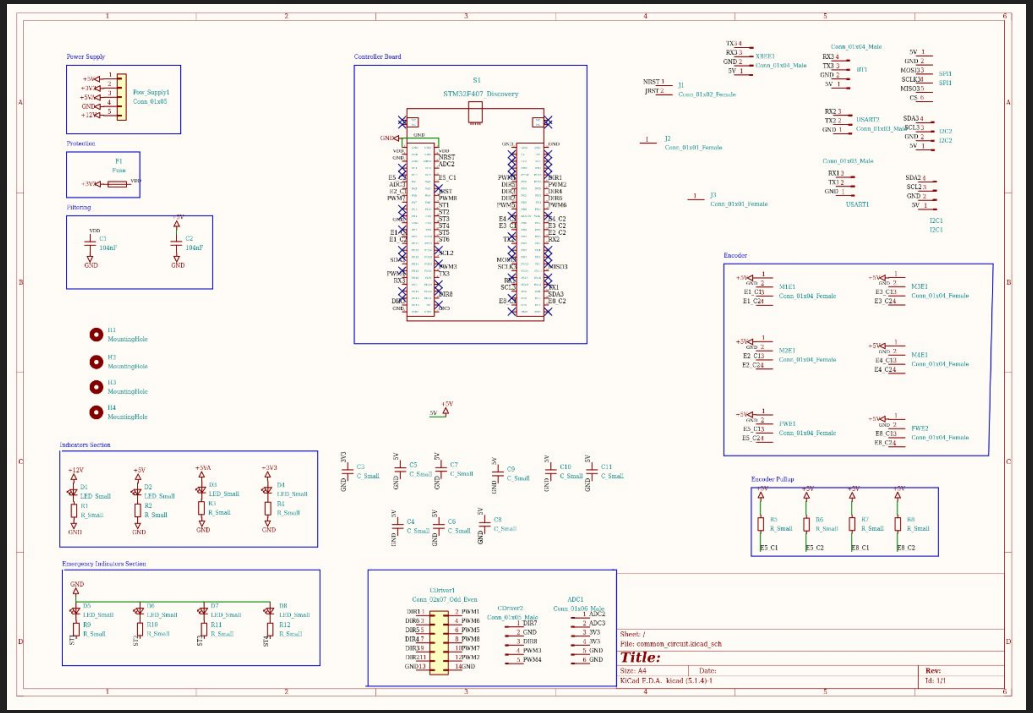


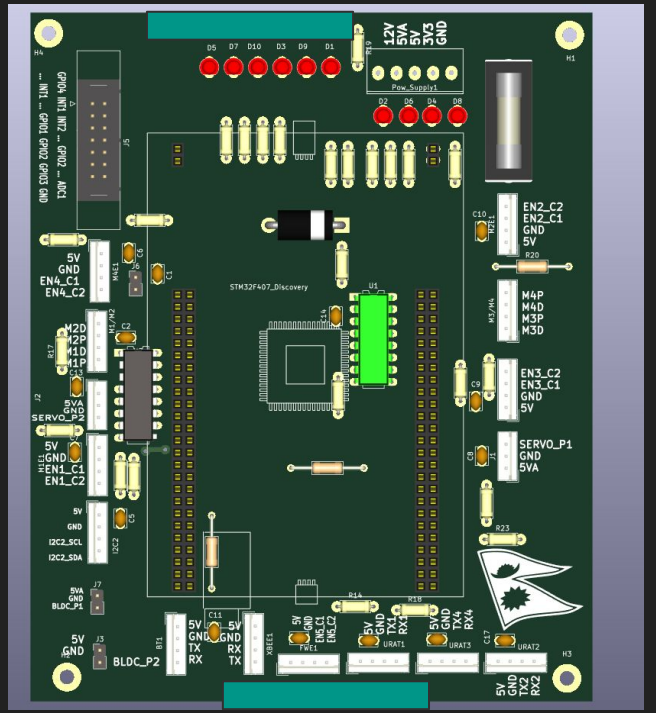
# Initial Circuit Design



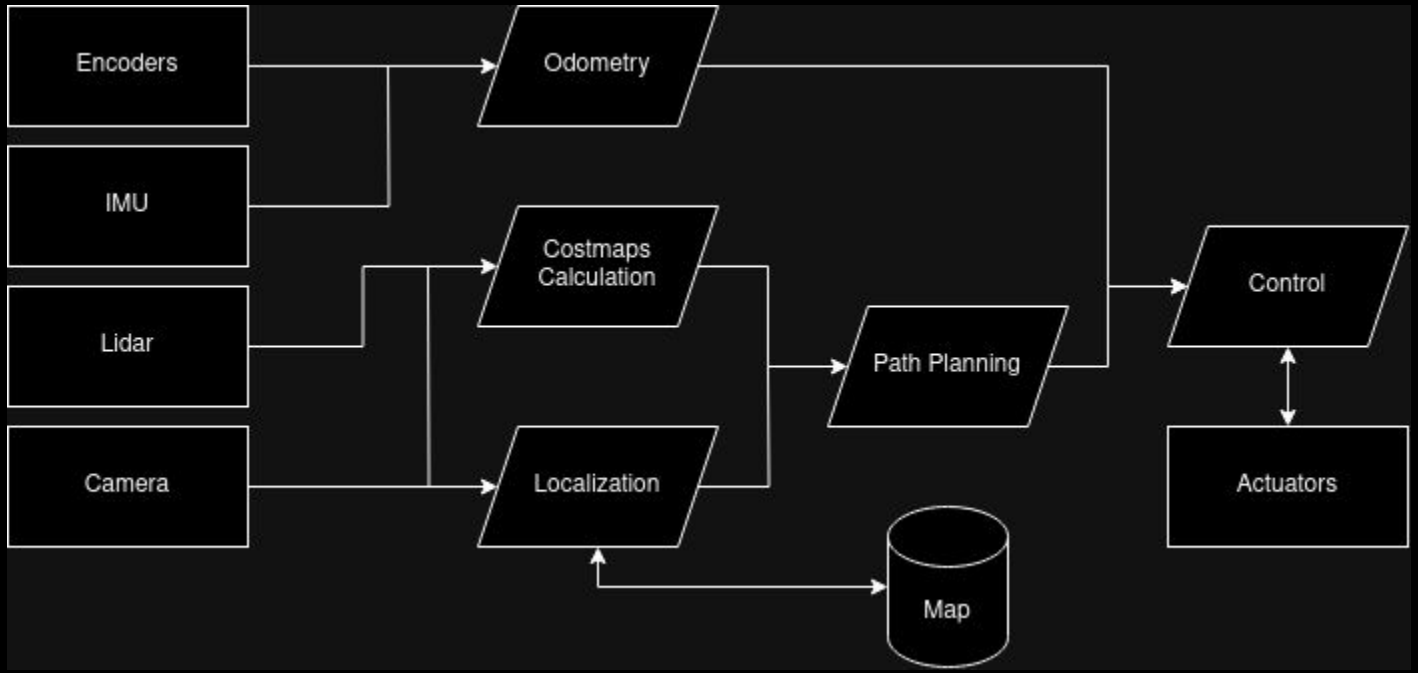
# Final Circuit



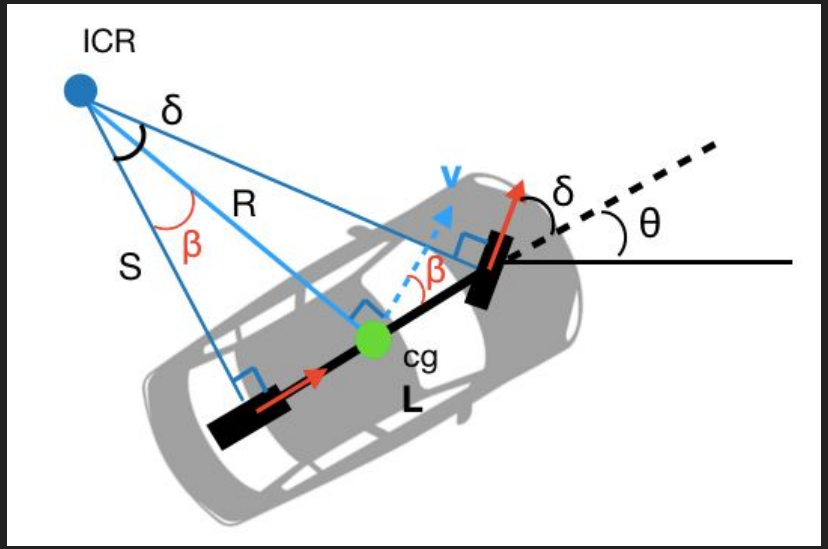
# Software Design

# System Design



# Control & Actuation

- UART bridge to Microcontroller to Raspberry Pi
- **bicycle steering model** for forward and inverse kinematics



# Position Estimate

Odometry
(linear and angular vel)

+

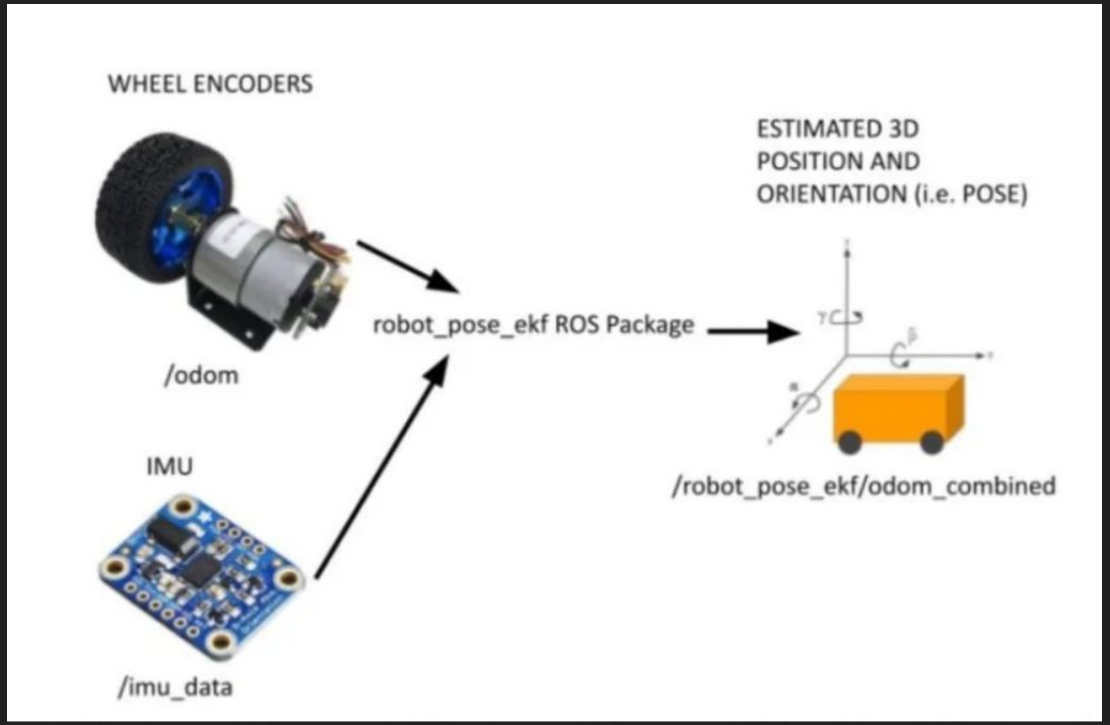
IMU
(linear accel + angular vel)

+

Position Estimate from SLAM

=

Final Position estimate



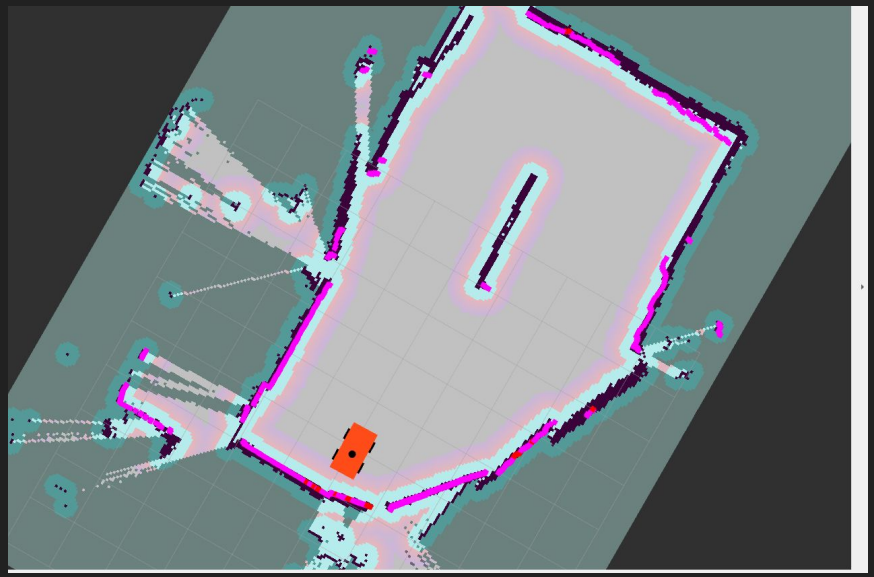
# Mapping

- Used slam_toolbox
- Car was driven manually in the environment to generate a map
- Localization of car in map was done using slam_toolbox and Particle Filter (AMCL)



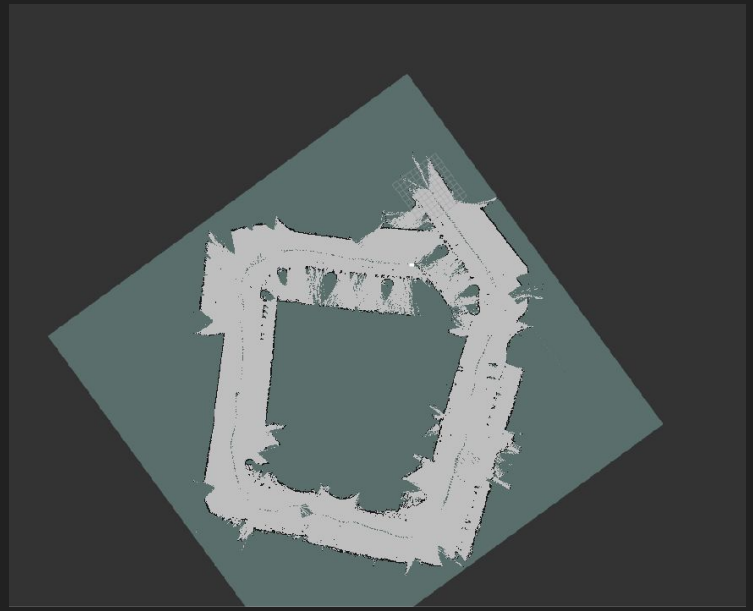
# Indoor Map





# Outdoor Map



15

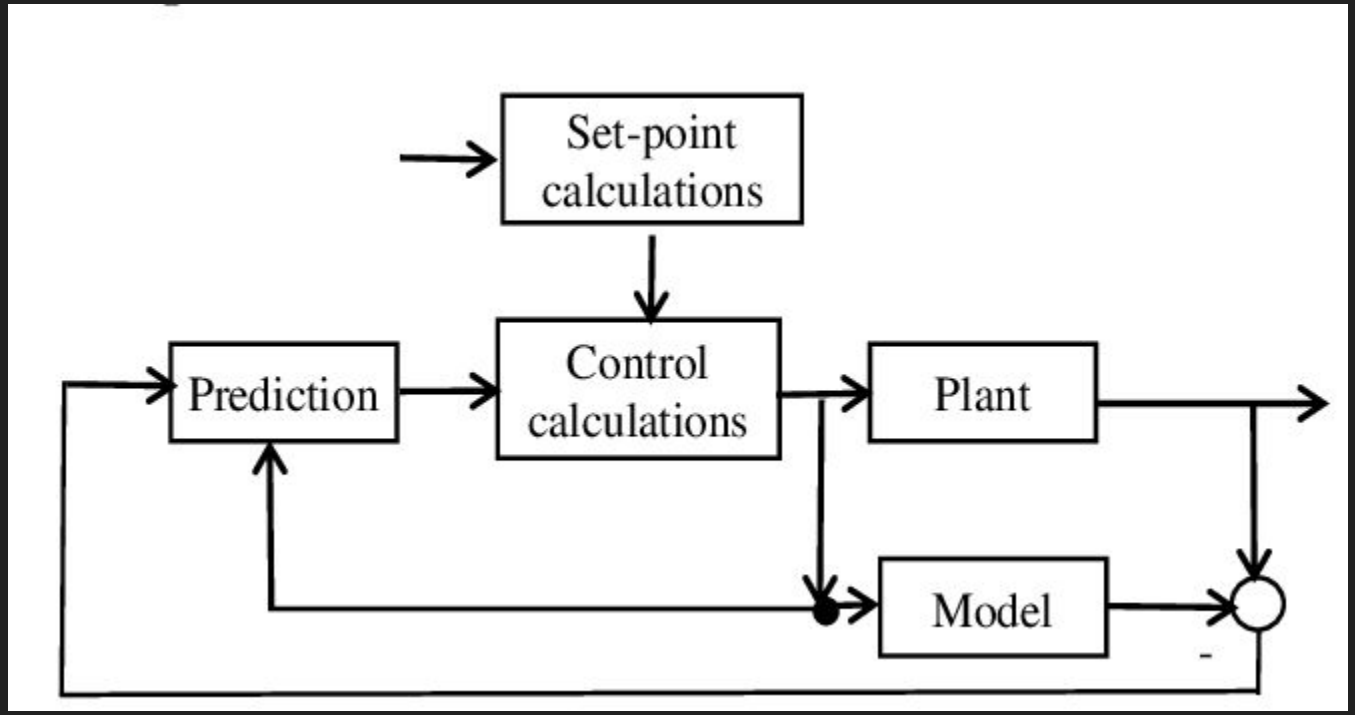


# Autonomous Navigation

- Based on Nav2 stack in ROS2
- Planner - Find the shortest path to goal
- Controller - Execute control actions to get to goal
- Behaviour Server - High level behaviors
  - Reroute when stuck
  - Go around persistent obstacles
- Smoothers
  - Smoothes the jagged path given by the planner
- Costmaps
  - Add factors like obstacles and road geometry

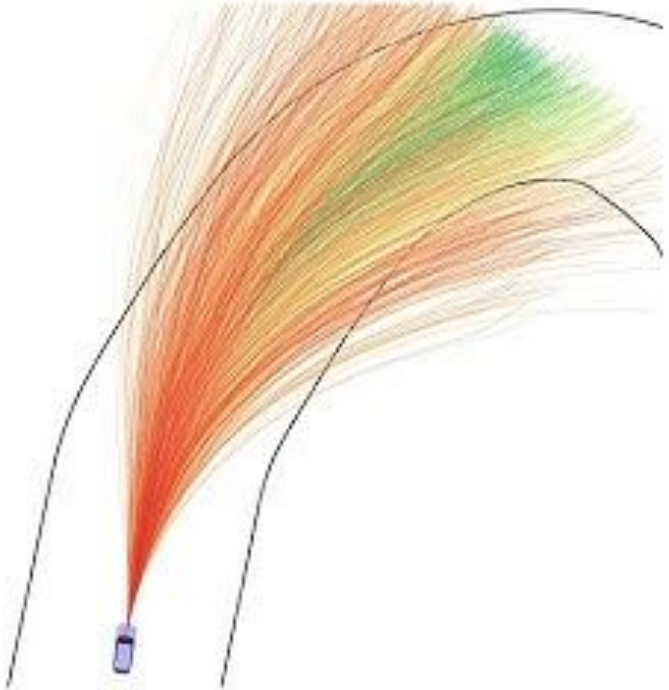
# Model Predictive Control

- Model Predictive Path Integral (MPPI) controller for tracking trajectories

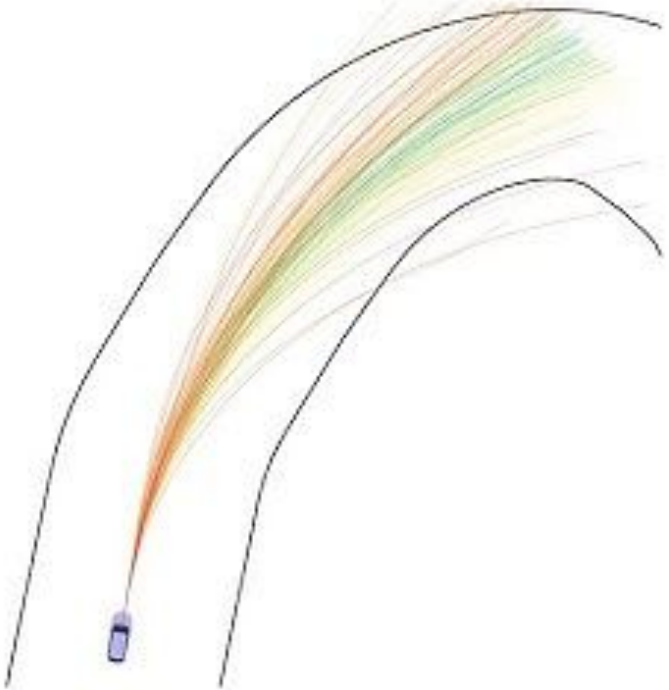


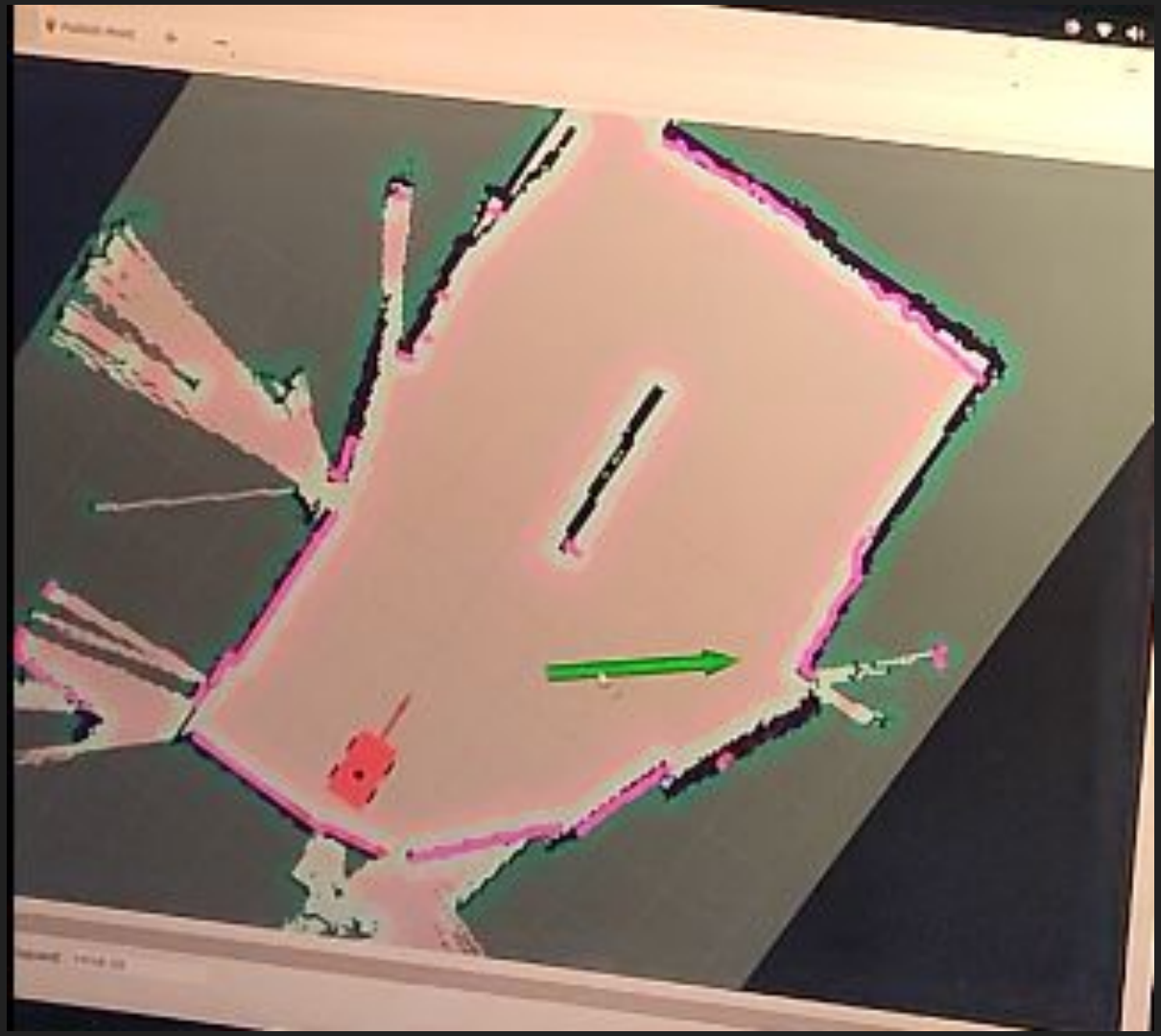
# Model Predictive Optimized Path Integral Control (MPOPI)

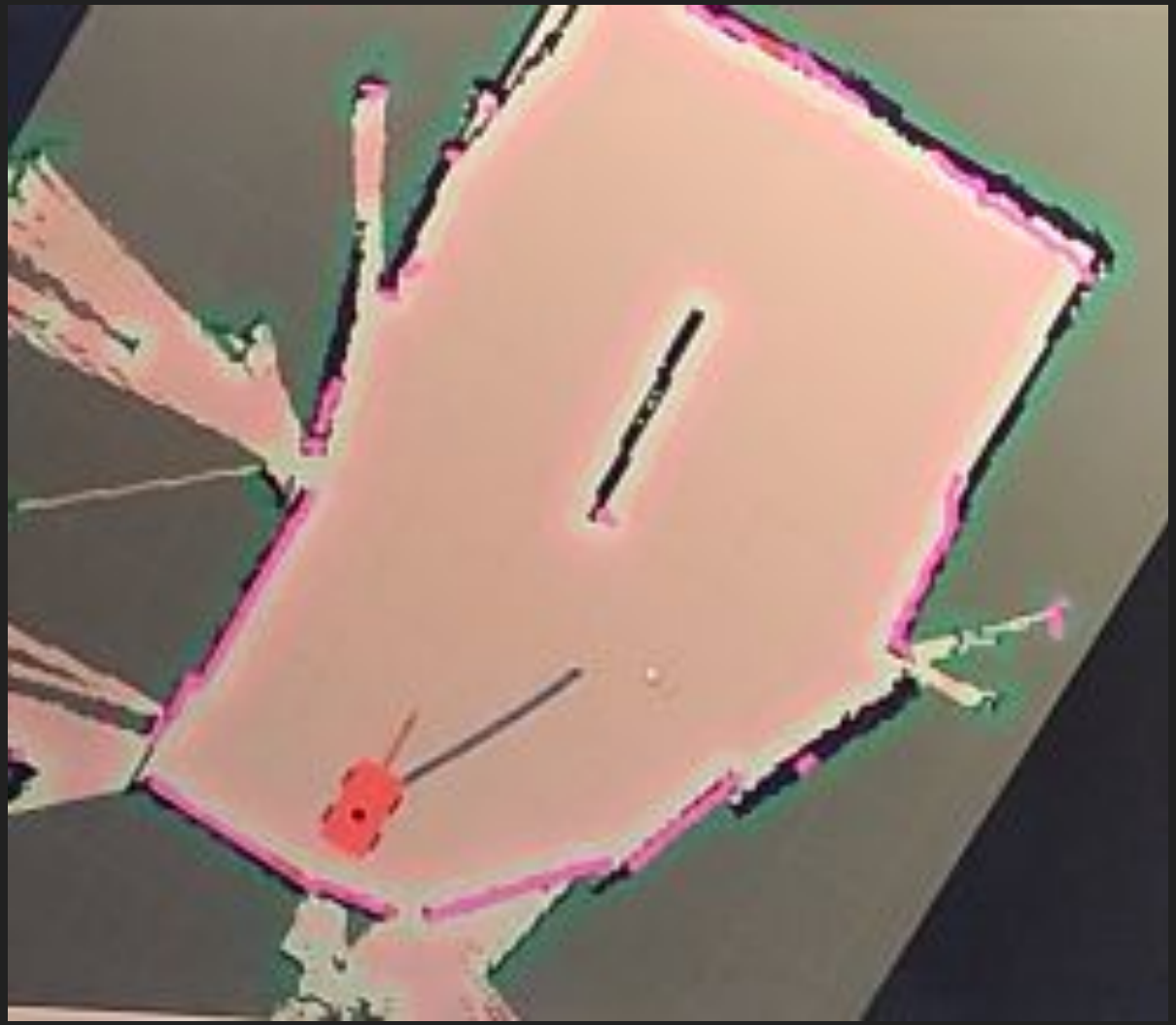
MPPI – 1500 Samples

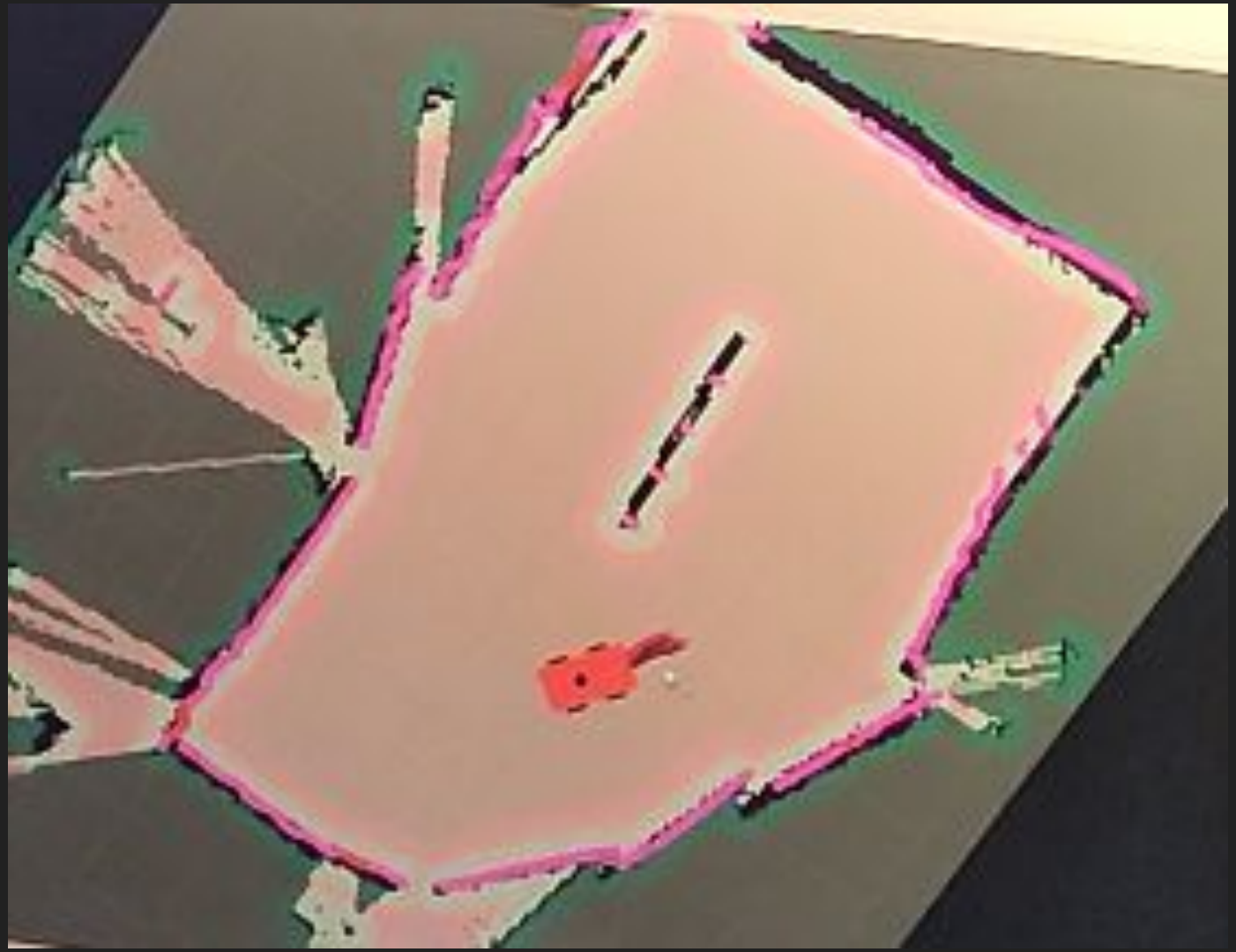


MPOPI (CE) – 150 Samples
Iteration 10



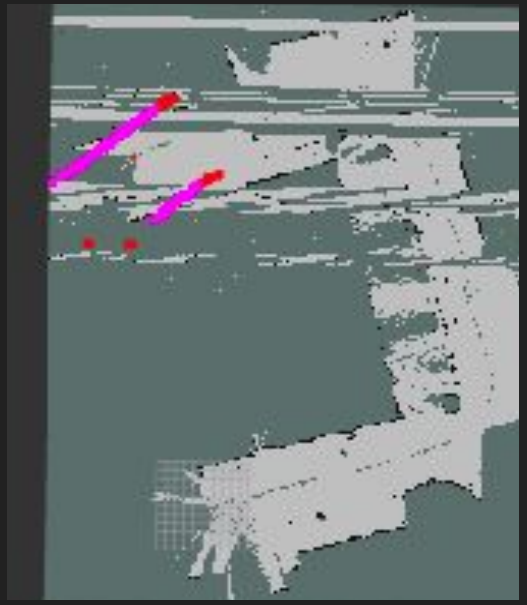




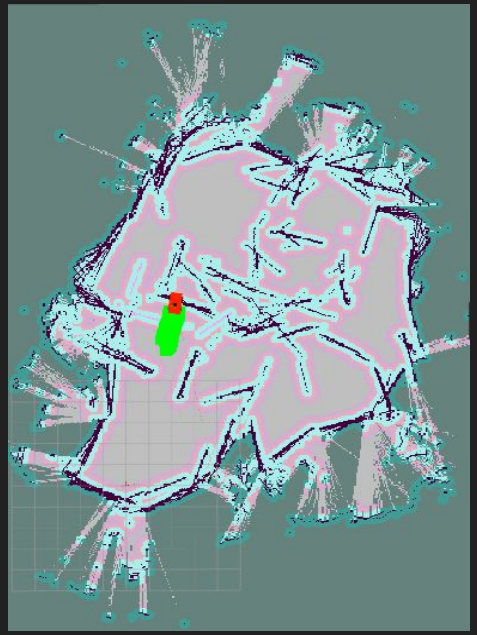


# Mapping Problems

Slow Network



Bad Odometry



# Visual Perception

# Dataset Curation



# Object Detection






# Image Segmentation

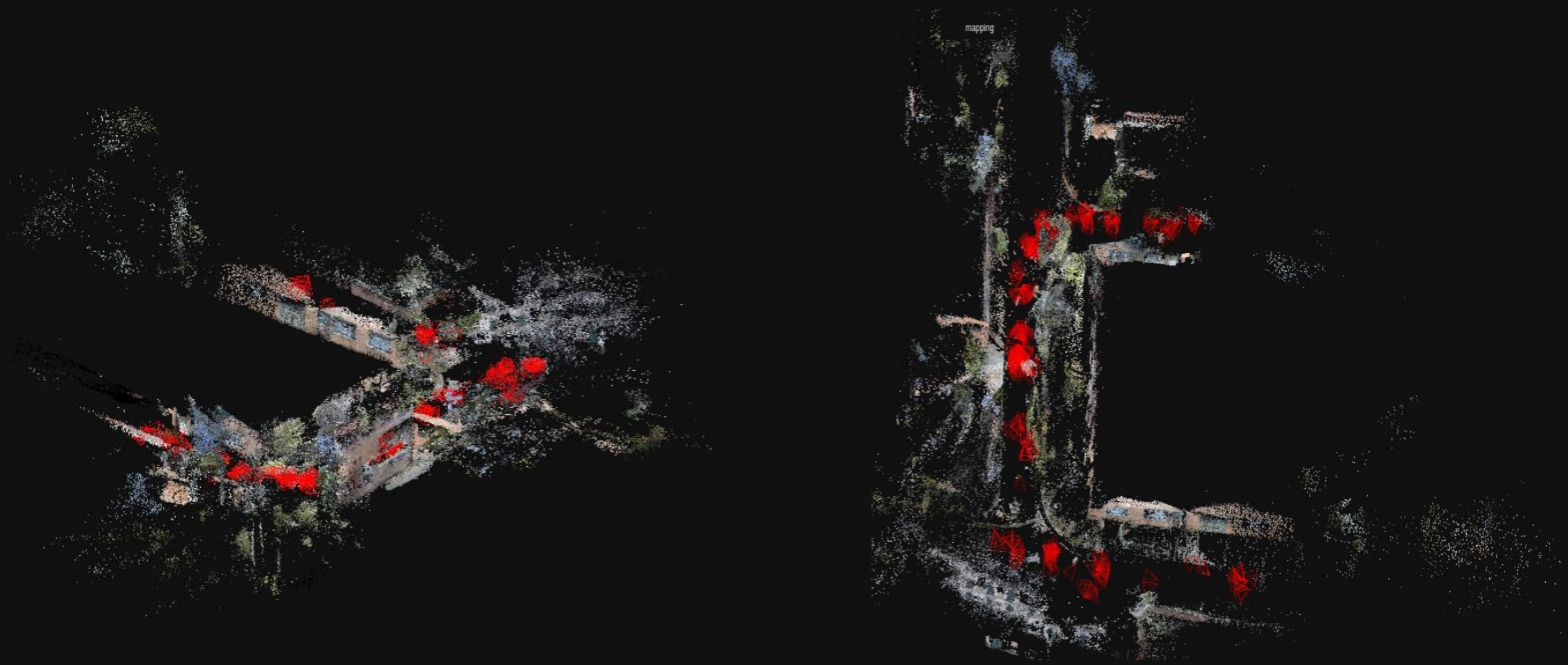


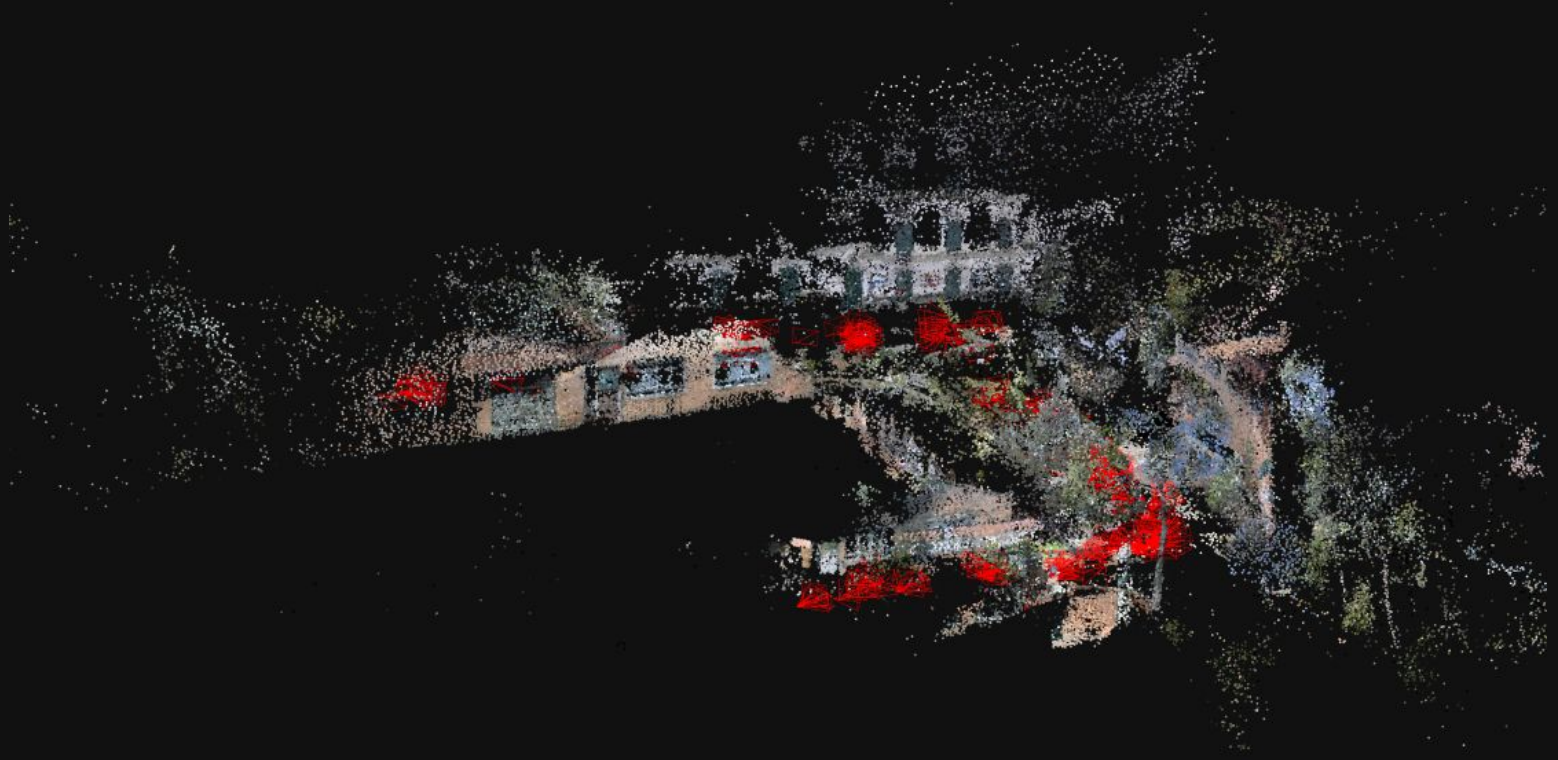






# Hierarchical Localization





# Conclusion

- Built a simple self driving car
- Used cheap, commodity hardware
- Built lidar and visual maps
- Implemented path planning and control
- Tested in near-ideal conditions
- Built an extensible software framework for future work



# Future Work

- Work with real hardware
- Address Nepal’s domain gap
- Build robust communication systems








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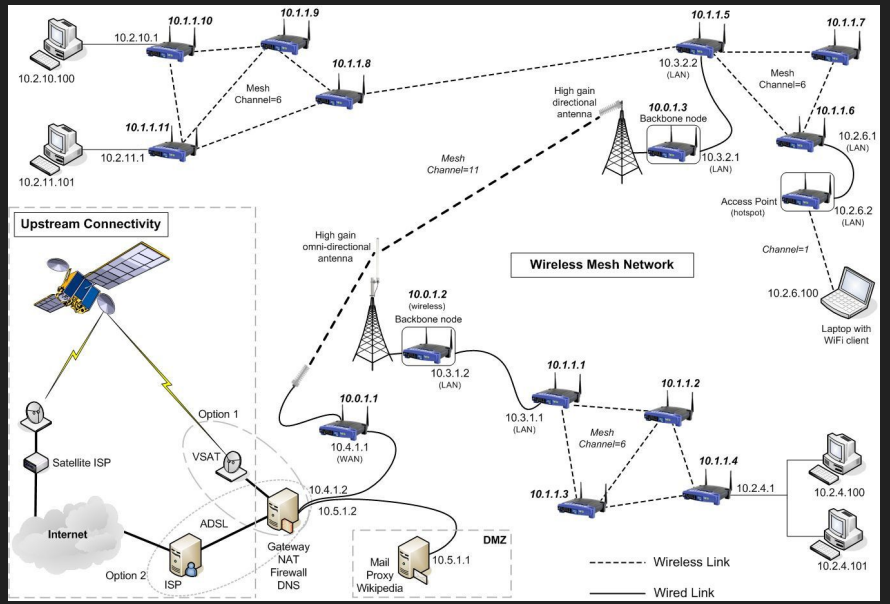






# Future Work

- Work with real hardware
- Address Nepal’s domain gap
- Build robust communication systems



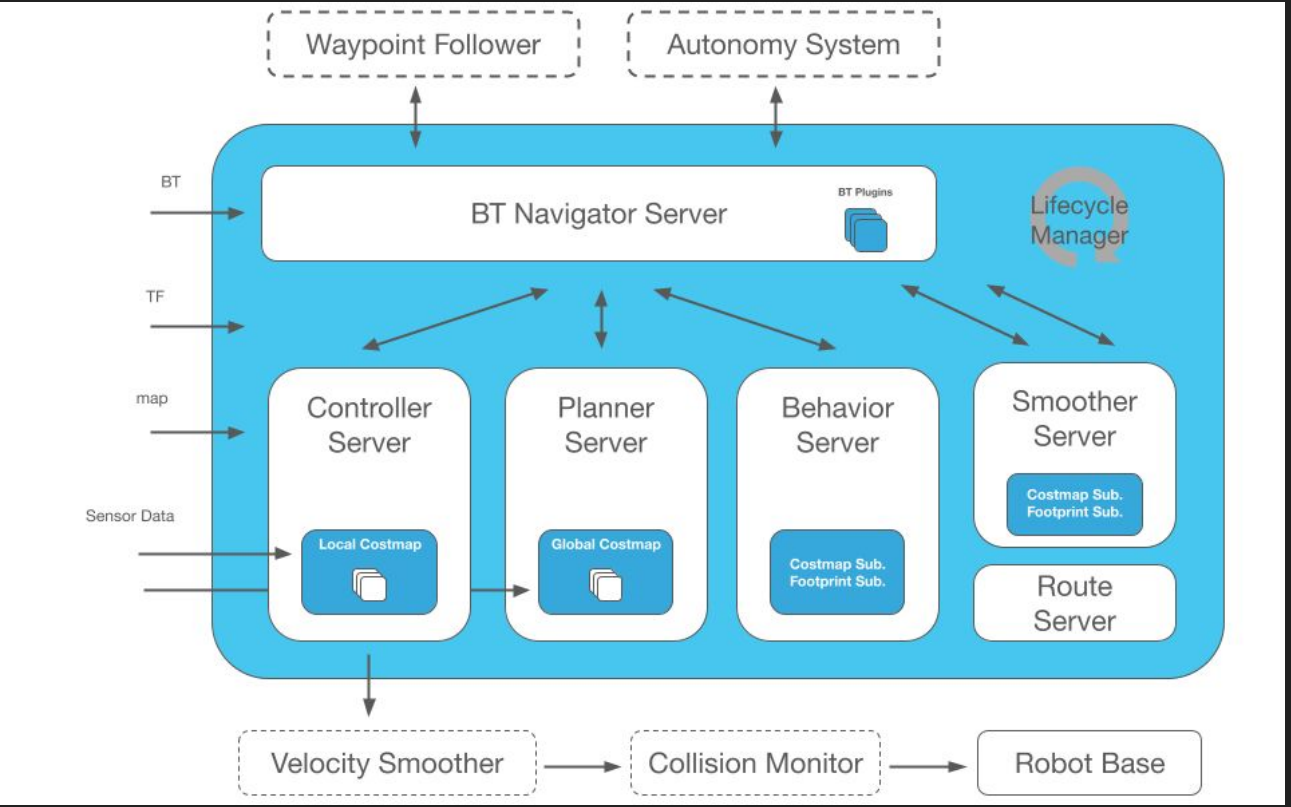
# THANK YOU

# Why MPPI Controller?

MPC variant that finds a control velocity for the robot using an iterative approach.

to adapt the controller to time-varying system dynamics

# nav2 Architecture

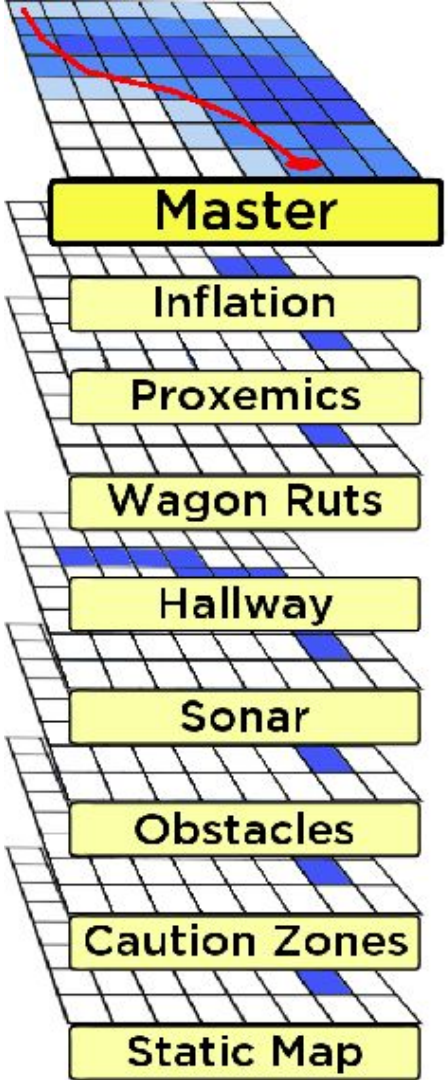


# ToC?

1. Introduction
    a. Why sdc? (safety, homegrown thing)
    b. Hami robotics club ko, this is relevant because:
        i. Robocon ma ros use gariyo but we wanted to build a real world robot with ROS
        ii. Krishbin dai haru ko project dekhera inspire bhaiyyo, eutai thau ma kaam garim
    c. scope
2. System design - hardware components ra block diagram of the system
    a. Distributed architecture, offload computation to more capable computers, kata bata kata kun protocol ma communicate bhairacha etc
    b. Circuit design? Use chai gariyena because ship bhayena time ma but still design dekhaune
    c. Car ko design etc
3. Control loop - software (car chalda chai ke ke chalirahuncha software ma)
    a. Nav2, control cascades, controller kata kun use bhairacha
4. Mapping and odometry (odom, ekf fusion, ulto odom ko kura etc, rati mapping gareko, localization ko pida)
5. Perception and localization (yolo objdet, segmentation and hloc)
6. Demo video?
7. Conclusion (goals achieved, future directions and stuff, limitations and gaps we found to work on in the future)
    a. Network is hard, ajhai garo cha robust comms

# Task Remaining

- Mapping
  - Visual+2D Lidar metric map
- Odometry
  - Fusion of odometry data from 4 sources
- Costmap factors and fusion
  - Obstacle avoidance + road keeping + lane preferences
- Perception Models
  - Tune obstacle detection and drivable region segmentation
- Navigation testing and tuning
  - End to end tests of planners, controllers, behaviour trees




Aggreagate layer, where path planning occurs.

Buffer zone around all obstacles

Detect people and respect their personal space

Go with the standard flow of traffic

Prefer to move down the right side of hallways

Probabilistic obstacles populated by sonar

Deterministic obstacles from lasers and other sensors

Avoid dangerous areas like kitchens when possible

Occupancy grid determined a priori

# Abhigyan’s notes

- They told us to talk to sdc major project ko software group, garenam, kei gareko jasto garna milcha? Maybe look at their reports for a general idea of what they are doing and say they were working with us club mai, closely kura kani bhairacha
- Segmentation, obj detection, hloc ko kura garne what we did bhanera, and then say unit chai banaim but didnt get time to integrate
- Focus big on control/odometry? Ros2_control ra ekf tira ekdam ramro kaam gareko ho arjan le, highlight garnu parcha
- Did mapping using slam
- Navigation task like to go from one point to another point